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Early Cambrian location and affinities of the Brunovistulian terrane (Central Europe) in the light of palaeomagnetic data

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Abstract: The Brunovistulian was one of the first tectonic units of Central Europe to be defined as a 'terrane'. In spite of extensive studies, the Early Palaeozoic palaeogeographical position and provenance of this unit remains unclear. A palaeomagnetic study of the Lower Cambrian red beds and a study of the trilobite fauna were performed to provide constraints on the palaeogeographical position of the terrane in Early Cambrian time. Good quality palaeomagnetic data obtained from the Lower Cambrian red beds suggest the mixed nature of the Early Cambrian geomagnetic field and a nearly equatorial position (a palaeolatitude of *c*. 7°) of the Brunovistulian terrane in the Early Cambrian. Comparison of this palaeolatitude with existing palaeogeographical models leads to the conclusion that at this time the Brunovistulian terrane was separated by a great distance from the Avalonian margin of Gondwana. The terrane was located within the Cadomian belt, occupying a position north of the present-day northern margin of Africa. It was coupled to the present-day southern margin of Baltica during the Cambrian, when Baltica moved along the Cadomian margin of Gondwana. Another possibility is that the Brunovistulian terrane could have remained near this margin of Baltica since Grenvillian time and was incorporated into the Cadomian orogen.

Keywords: Brunovistulian terrane, Cambrian, palaeogeography, palaeomagnetism, trilobite fauna.

The Brunovistulian terrane (e.g. Dudek 1980; Pożaryski 1991; Dadlez 1995; Buła *et al*. 1997) is situated near the southwestern edge of the East European Craton, in the area of the Trans-European Suture Zone (Fig. 1). It was one of the first tectonic units of Central Europe to be defined as a 'terrane' (Brochwicz-Lewiński *et al*. 1986) and still remains the best documented terrane of the central part of the Trans-European Suture Zone (Pożaryski 1991; Dadlez 1995; Franke 1995; Pharaoh 1999). An exotic position of the Brunovistulian terrane with respect to the East European Craton, as well as to the Małopolska and Bohemian Massifs during the early Palaeozoic, is suggested by the composition of the basement (Dudek 1980), the development of sedimentary cover (Buła *et al*. 1997; Buła 2000), and the presence of major wrench zones, which border the Brunovistulian terrane, particularly in the north and east (Żaba 1999; Buła 2000). The researchers postulating an allochthonous origin for the Brunovistulian terrane interpreted it as a fragment of the crust derived from the Gondwanan margin. This was based on the general geological framework (Unrug *et al*. 1999), faunal provincialism (Moczydłowska 1997), the results of provenance studies of detrital minerals (Belka *et al*. 2000), Nd model ages and the age of single zircon grains from the basement (Finger *et al*. 2000*a*, *b*; Hegner & Kröner 2000; Friedl *et al*. 2001). According to these interpretations of the Brunovistulian terrane as an allochthonous terrane, it was located in the peri-Gondwanan belt during the Neoproterozoic–Early Cambrian, in its South American (Hegner & Kröner 2000; Friedl *et al*. 2001) or North African region (Unrug *et al*. 1999; Leichman & Höck 2001). A more stationary model was presented by Żelaźniewicz (1998) and Żelaźniewicz *et al*. (2001), who suggested, on the basis of U–Pb sensitive high-resolution ion microprobe (SHRIMP) data, that the Neoproterozoic (Vendian) foreland flysch basin of the Brunovistulian terrane and neighbouring Małopolska block was developed next to the East European Craton.

The basement of the Brunovistulian terrane is composed of metamorphic and igneous rocks, mostly Cadomian (Neoproterozoic) in age (Dudek 1980; Van Breemen *et al*. 1982; Finger *et al*. 2000*a*, *b*; Żelaźniewicz *et al*. 2001). In some places they are covered by Vendian greenschist-grade metasediments (Żelaźniewicz *et al*. 2001). The Cadomian basement of the Brunovistulian terrane is overlain by gently deformed Lower Cambrian to Ordovician unmetamorphosed sedimentary rocks (Buła 2000). The chronostratigraphy of the Lower Cambrian clastic sediments, which in the northern part of the Brunovistulian terrane continue into the Middle Cambrian and Ordovician units, is based on acritarch and trilobite fauna (Orłowski 1975; Moczydłowska 1995, 1997; Jachowicz & Přichystal 1998). These sediments are subdivided into two lithostratigraphic units: an older Borzęta Formation and a younger Goczałkowice Formation (Fig. 1c) (Buła *et al*. 1997). The sediments of the Borzęta Formation are recognized only along the eastern margin of the Brunovistulian terrane, with a maximum thickness of about 1000 m. The thickness of the Goczałkowice Formation increases from the SW to the east and north, and is estimated to be up to *c*. 2000 m in the marginal part of the Brunovistulian terrane. The basal sequence of the Cambrian, i.e. the Mogilany Member of the Goczałkowice Formation, consists of hematitic quartzo-feldspathic sandstones, and pebbly sandstones. The remaining part of the Cambrian sequence represents a complete marine transgressive cycle, which follows terrestrial deposition of the basal sequence (Moczydłowska 1997).

The Lower Palaeozoic rocks of the Brunovistulian terrane are covered by Lower Devonian sandstones of the 'old red' type (Konior & Turnau 1973). They were succeeded by the development of thick carbonate cover that accumulated up to Viséan time (Narkiewicz 1978; Belka 1985), followed by Viséan flysch sediments in the western part of the Brunovistulian terrane

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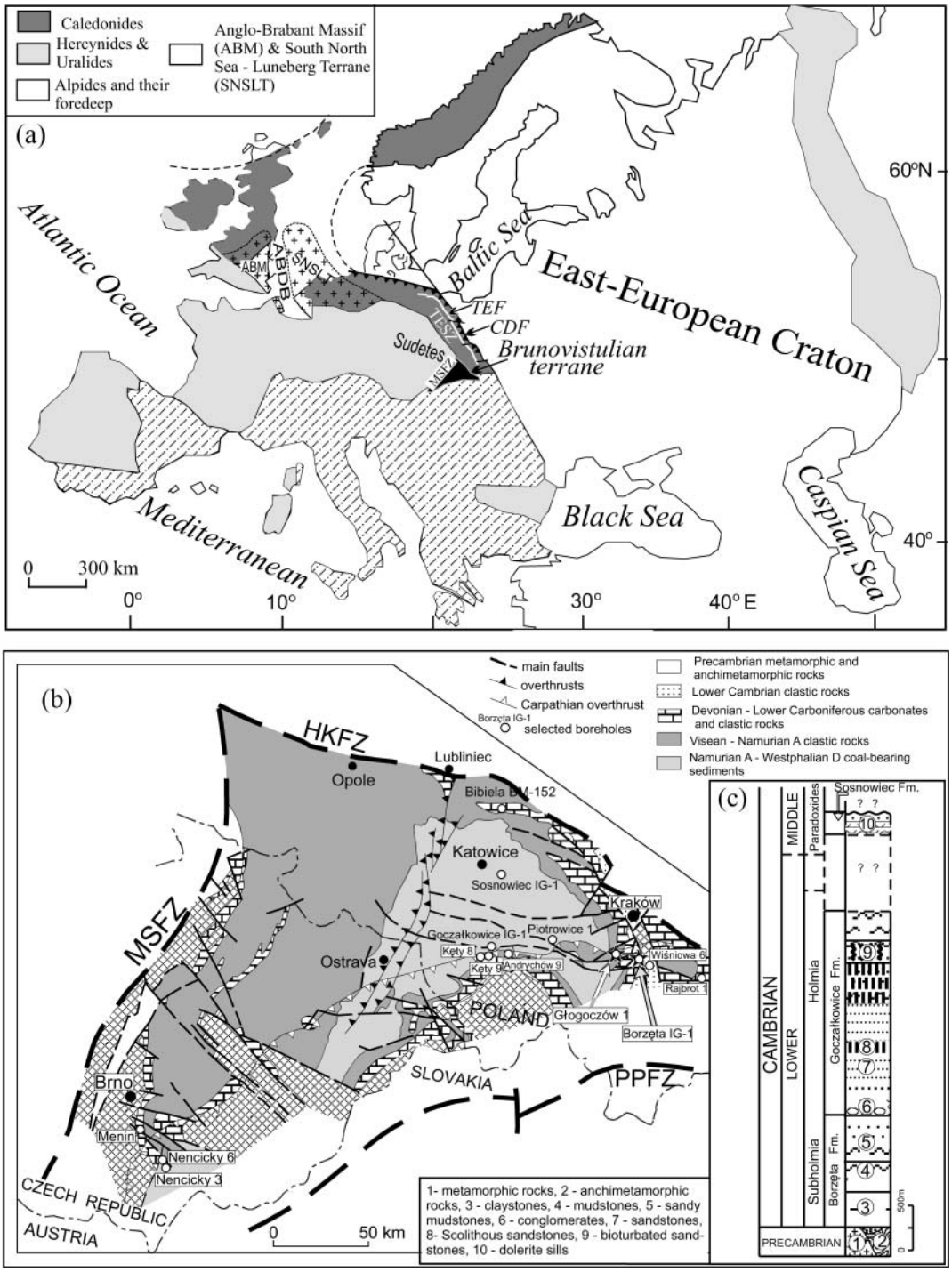


Fig. 1. (**a**) The location of the Brunovistulian terrane within a tectonic sketch map of Europe. TESZ, Trans-European Suture Zone; TEF, Trans-European Fault; CDF, Caledonian Deformation Front; MSZ, Moravo-Silesian Zone; ABDB, Anglo-Brabant Deformation Belt. (**b**) Simplified geological sketch map of the Brunovistulian terrane. MSFZ, Moravo-Silesian Fault Zone; HKFZ, Hamburg–Krakow Fault Zone; PPFZ, Peri-Pieniny Fault Zone. (c) Stratigraphic subdivision of Cambrian sediments from the Brunovistulian terrane.

(Belka 1987). In the early Namurian this was replaced by paralic and subsequently limnic coal-bearing molasse. This stage of the basin development is often considered as a Variscan foredeep (e.g. Kotas 1990).

The discrepancies between the models of palaeogeographical position of the Brunovistulian terrane partly result from lack of palaeomagnetic and tectonic constraints. Also, the Cambrian trilobite faunas, first described about 30 years ago, require verification. The aim of the present paper is to address these deficiencies. The opportunity to do this is due to the existence of several deep boreholes in the unmetamorphosed Cambrian cover.

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Palaeomagnetic results

Sampling and palaeomagnetic procedure

A total of 25 Early Cambrian fragments of drill cores (up to 40 cm long) were collected for palaeomagnetic investigations from two vertical boreholes: Borzęta IG-1 and Glogoczów IG-1, situated in the eastern part of the Brunovistulian terrane (Fig. 1). The deviation of wells from vertical was less than 3° in the intervals of sampling. This deviation was much larger (up to 27°) only in the lowest (not sampled) part of Borzęta IG-1 borehole. Samples were oriented top–bottom and their horizontal orientation was determined according to the regional direction of dip inferred from borehole data and seismic sections crossing the sampled boreholes. These rocks developed as typical red beds, and dip 10–18° towards the west (264° of azimuth of dip calculated from seismic data). Their stratigraphic age is close to the *Holmia–Subholmia* boundary. *Holmia* trilobites were found (Biernat & Baliński 1973; Orłowski 1975) in the uppermost Goczałkowice Formation (not occurring in the studied sections) and acritarcha of *Subholmia* age were recognized in the middle part of the Borzęta Formation (Buła *et al*. 1997; Buła 2000). Hence the numerical age of the studied rocks should correspond to *c*. 535 Ma (Gradstein & Ogg 1996). The orange colour of most specimens of acritarcha points to the low degree of thermal maturation of the studied rocks.

Several core specimens 2.5 cm in diameter and 2.2 cm in length were drilled from each fragment of the drill core. Pilot specimens were subjected to both thermal and alternating field demagnetization experiments. Thermal demagnetization was much more effective and most of the specimens were demagnetized using this technique. Demagnetization results were analysed using orthogonal vector plots (Zijderveld 1967), and the directions of the linear segments were calculated using principal component analysis (Kirschvink 1980). Magnetic mineralogy was determined from isothermal remanent magnetization (IRM) techniques and thermomagnetic analyses (Lowrie 1990). Anisotropy of magnetic susceptibility (AMS) was measured on a KLY-2 susceptibility bridge, and the resulting AMS fabrics were computed using the aniso program (Jelinek 1977).

Magnetic carriers and magnetic susceptibility anisotropy

IRM acquisition experiments and subsequent thermal demagnetization of the three-axis IRM acquired in fields of 0.1, 0.4 and 1.5 T (Fig. 2) indicate the presence of a high-coercivity mineral with unblocking temperatures higher than 600 °C. This suggests that hematite is the main remanence carrier in the studied red beds. Thermomagnetic curves do not indicate the presence of any other magnetic minerals. Moreover, the subordinate admixture of lower-coercivity phases seems to be connected with hematite as well. Measurements of the AMS revealed a low degree of magnetic grain alignment (up to 5%) with strong predominance of foliation (shape parameter up to 0.8). This foliation is bedding parallel (Fig. 3).

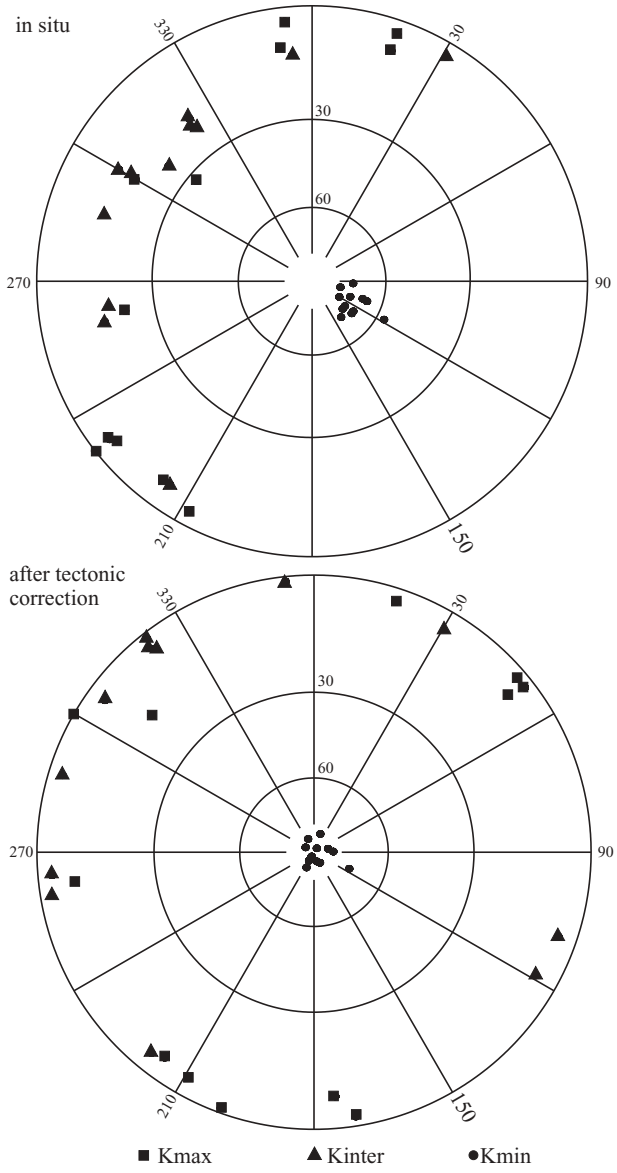


Fig. 3. Directions of magnetic anisotropy in the Lower Cambrian red beds from Borzęta IG-1 borehole.

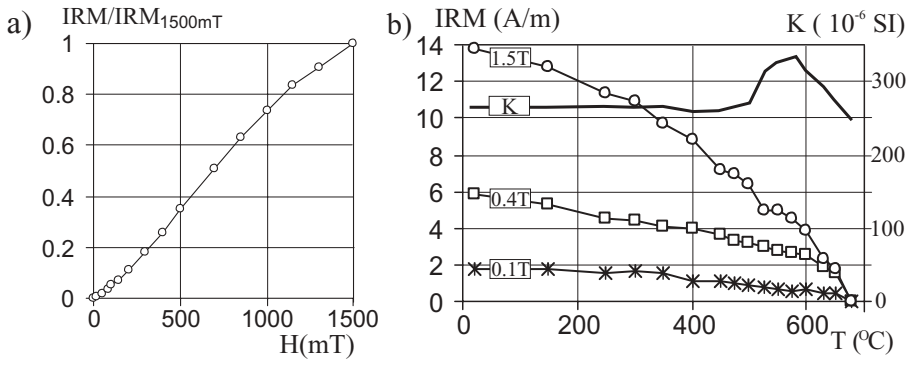


Fig. 2. (**a**) IRM acquisition curve for the sample of Cambrian red mudstone from Borzęta IG-1 borehole. (**b**) Thermal demagnetization of orthogonal-axis IRM curves obtained for the same sample.

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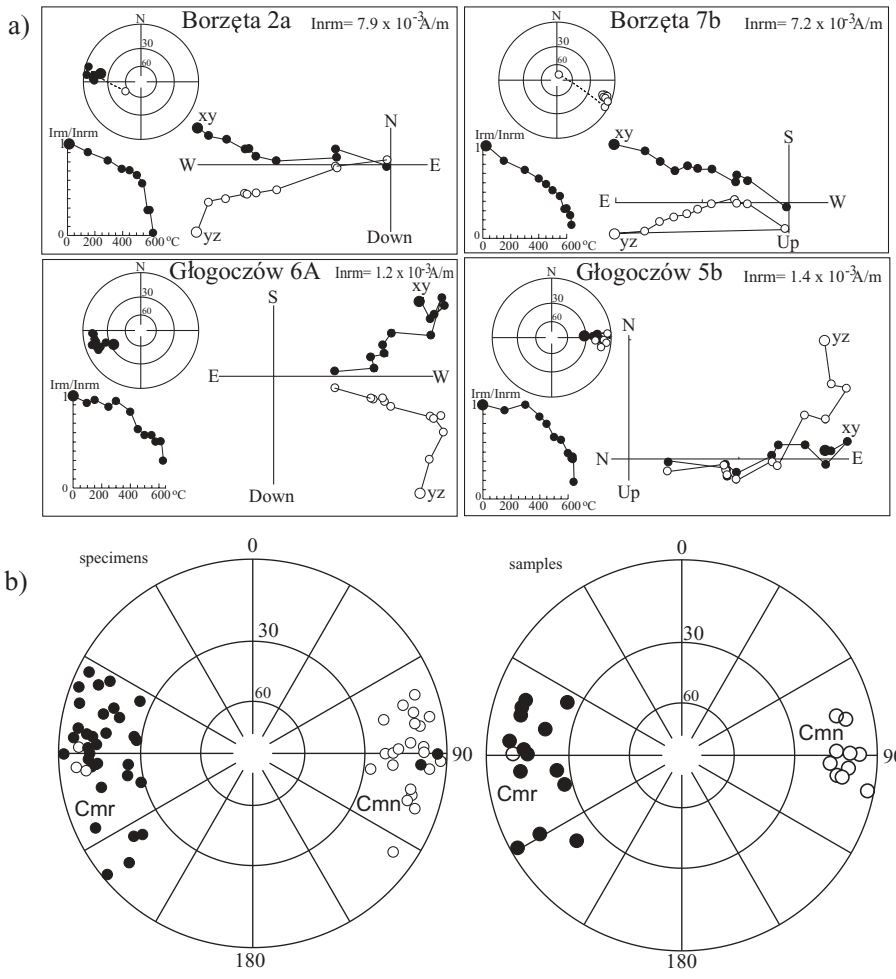
Results of thermal demagnetization

The natural remanent magnetization intensities of the studied red beds range between 0.9 and 9.5 mA m^{-1}. They were usually higher in the specimens from Borzęta IG-1 borehole. Thermal demagnetization revealed the presence of one distinct and well-clustered dual polarity component with shallow inclination and unblocking temperatures between 600 and 650 °C (Fig. 4). Some samples revealed a steep low-temperature natural remanent magnetization (NRM) component removed by temperatures higher than 150 °C. This component, probably of viscous origin, was strongly dispersed and therefore was not evaluated statistically. The unblocking temperatures indicate that hematite is the carrier of dual-polarity component. It was assumed that directions with eastern declinations represent the normal polarity geomagnetic field. The observed mixed polarity pattern is not chaotic. Several successive samples have the same polarity (Fig. 5). The mean inclination on the sample level (14°) indicates a palaeolatitude of 7°.

Trilobite fauna

Cambrian trilobites were found (Biernat & Baliński 1973; Orłowski 1975) only in the Goczałkowice IG-1 borehole (Fig. 1). They occur between 2793 m and 2850.45 m within grey–greenish siltstones with rare intercalations of fine-grained sandstones, representing the Goczałkowice Formation. Trilobite remains are represented by 11 fragmentary specimens, dominated by cranidia. Orłowski (1975) noted the presence of three species: *Schmidtiellus panowi* (Samsonowicz), *Strenuaeva primaeva* (Brøgger) and *Ellipsocephalus nordenskjoeldi* Linnarsson. However, during our revision of the fauna, only the presence of *Schmidtiellus panowi* (Samsonowicz) has been confirmed. It is represented by one incomplete cranidium (Fig. 6). The remainder of the trilobites, although more abundant in the analysed section, are rather poorly preserved and have been assigned only at genus level. They include *Strenuaeva* sp. and *Ornamentaspis* sp.

Schmidtiellus panowi (Samsonowicz) is known solely from



c)

	N	n	D	I	A_{95}	K	PLAT	PLONG	dp	dm	LAT
Cmr		36	269	11	6.6	15.3					
Cmn		26	91	- 10	5.3	29.5					
Cm		62	90	- 11	4.2	20.1	4°N	293°E	2	4	5°S
Cmr	14		267	15	10.5	15.3					
Cmn	11		89	- 12	6.5	50.3					
Cm	25		88	- 14	6.2	22.6	4°N	296°E	3	6	7°S

Fig. 4. (a) Typical demagnetization characteristics (demagnetization directional tracks, intensity decay curves and orthogonal plots) of Lower Cambrian red beds from Borzęta IG-1 and Glogoczów IG-1 boreholes. (**b**) Stereographic projections (at specimen and sample level) of high-temperature line-fit palaeomagnetic directions from the Lower Cambrian red beds drilled in Borzęta IG-1 and Glogoczów IG-1 boreholes. (**c**) Mean directions and palaeopoles for these normal and reversed polarity directions. In the orthogonal plots: ○, vertical projections; ●, horizontal projections. In the stereographic projections: ○, upward pointing inclinations; ●, downward pointing inclinations. The orthogonal and stereographic plots are presented in tilt-corrected coordinates. N, number of samples used in calculations; n, number of specimens used in calculations; D/I, declination/inclination (in degrees); K, precision parameter (after Fisher 1953); A_{95}, semi-angle of the cone of 95% confidence; PLAT, latitude of south palaeomagnetic pole; PLONG, longitude of south palaeomagnetic pole; dp, error of the distance between site and palaeopole; dm, palaeodeclination error; LAT, palaeolatitude of studied area.

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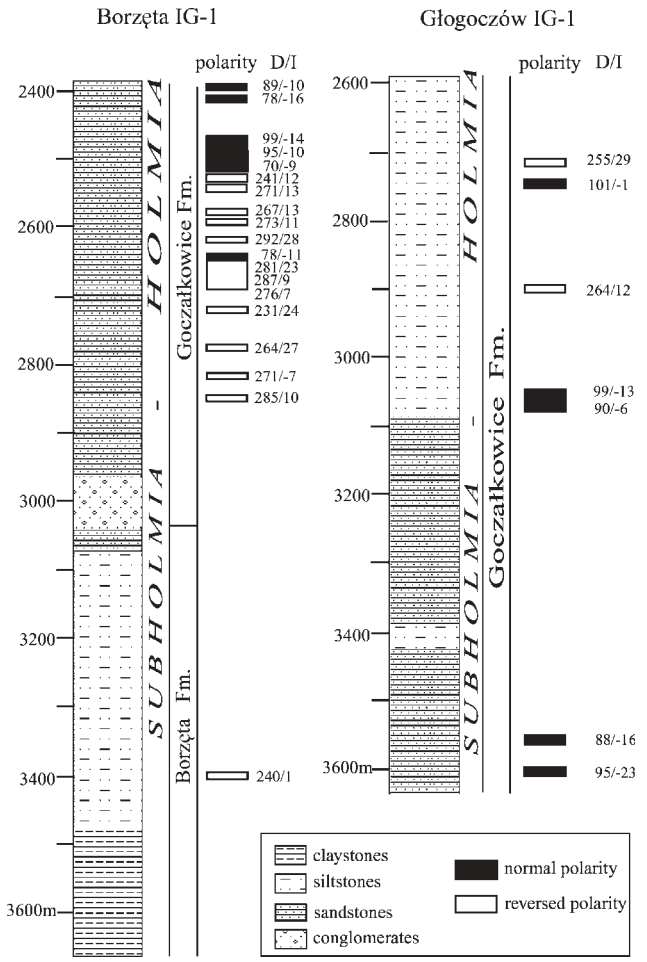


Fig. 5. Magnetic polarity changes in the Lower Cambrian red beds drilled in Borzęta IG-1 and Glogoczów IG-1 boreholes.

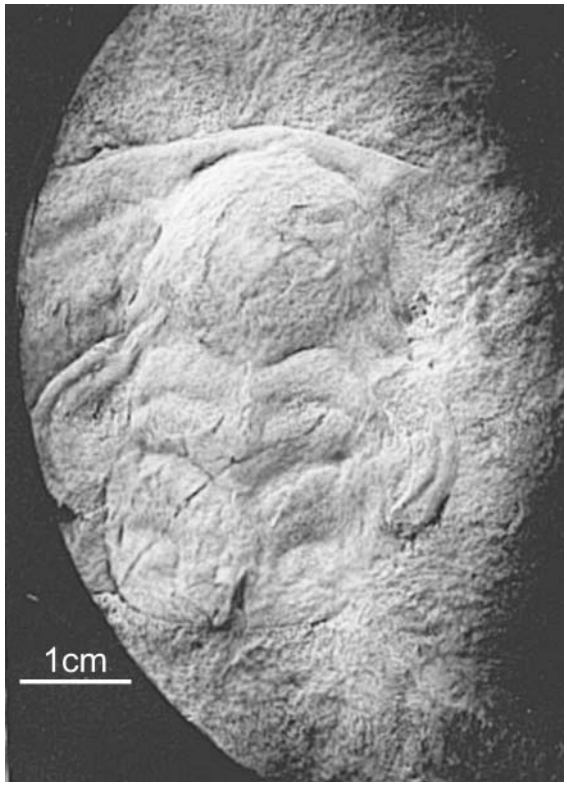


Fig. 6. Early Cambrian trilobite *Schmidtiellus panowi* (Samsonowicz) from the Goczałkowice IG-1 bore core. This trilobite is common for the Brunovistulian and Małopolska terranes.

Poland, where in addition to the Silesian area it occurs in the southern (Kielce) region of the Holy Cross Mountains (Orłowski 1985). *Schmidtiellus* is characteristic of the palaeocontinent of Baltica, being recognized in Lower Cambrian strata of Scania (Sweden) and Estonia (Bergström 1973). *Strenuaeva* and *Ornamentaspis* are not palaeogeographically indicative. *Ornamentaspis* Geyer, characteristic of the Lower Cambrian strata, is widely distributed, being represented in Scandinavia, Estonia, the Bohemian Massif, Spain and Morocco (Geyer 1990). *Strenuaeva* Richter & Richter, considered to be a Baltic genus, was also recognized recently in Moroccan sections (Geyer 1990).

The biogeographical affinities of the fauna are thus not clear. Affinities to Baltica were often proposed (Orłowski 1975; Buła *et al*. 1997; see also Valverde-Vaquero *et al*. 2000), particularly based on the presence of a supposedly Baltic species *Schmidtiellus panowi* (Samsonowicz). This species is, however, so far considered endemic to the Małopolska and Upper Silesia blocks. Moreover, even though the Early Cambrian fauna of the Małopolska Block seems to be Baltic in character, it is dominated by species confined to this terrane (Orłowski 1985), suggesting some kind of isolation of the area (Żylińska 2002). Thus the endemic character of the only determined species and the recognition of the remained fauna at only the genus level (*Ornamentaspis* sp. and *Strenuaeva* sp.) are insufficient data to permit a unequivocal palaeogeographical solution.

Discussion

The characteristic component of magnetization of the Lower Cambrian red beds from the Borzęta IG-1 and Glogoczów IG-1 bore cores is most probably nearly primary. It records a mixed polarity pattern that corresponds to the mixed nature of the Early Cambrian geomagnetic field (Kirschvink *et al*. 1991; Pisarevsky *et al*. 1997). Its carrier, hematite, is confined to particular Cambrian members and the reddish coloration does not cross the stratigraphic boundaries. It is, therefore, most probably not a late diagenetic feature. An early diagenetic origin of hematite, however, cannot be excluded because there were no notable differences of characteristic inclination values in sandstones and mudstones. The uppermost member of the Lower Cambrian and the Middle Cambrian Sosnowiec Formation are entirely grey in colour (Buła 2000). The likely nearly primary origin of the remanence means that its age is *c*. 535 Ma. This age is the same as that of the Early Cambrian palaeomagnetic pole for Baltica (Torsvik & Rehnström 2001) and therefore these palaeomagnetic data could be used in a coeval palaeogeographical reconstruction. The palaeomagnetic data of Torsvik & Rehnström (2001) show the northern edge of Baltica (i.e. its present SW margin) near a latitude of 35° S. However, new good-quality palaeomagnetic data from the Vendian sediments of Baltica (Popov *et al*. 2002) point to the necessity of substantial revision of the Vendian–Cambrian segment of the Baltic apparent polar wander path (APWP). The APWP of Popov *et al*. (2002) indicates a more equatorial position of Baltica in the Early Cambrian than was inferred from the previous data. The closeness of the Late Vendian, Early Cambrian and Late Cambrian poles to Scandina-

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via may indicate their secondary (i.e. Late Cambrian) origin. The palaeomagnetic pole from this paper is very close to this revised Vendian–Cambrian segment of the Baltic APWP (Fig. 7).

The biogeographical affinities of the Early Cambrian trilobite faunas from the Brunovistulian terrane are not clear. Recognition of these faunas at genus level revealed the presence of taxa known from Gondwana and Baltica (*Ornamentaspis*, *Strenuaeva*) or Baltica only (*Schmidtiellus*). The endemic taxon *Schmidtiellus panowi* (Samsonowicz) has been recognized in the Małopolska Block and the Brunovistulian terrane only and therefore can point to the Early Cambrian proximity of both units. On the other hand, the Gondwanan and Baltic affinities of *Ornamentaspis* sp. and *Strenuaeva* sp. do not support a significant separation between the shelves of Baltica and Gondwana in the Early Cambrian.

The significance of acritarchs from the Brunovistulian terrane as a palaeogeographical indicator is even more ambiguous. Moczydłowska (1995) recognized associations taxonomically comparable with those recorded in Baltica, Laurentia and Gondwana, with the closest affinity to Iberia. Jachowicz & Přichystal (1998) pointed to similarities of the Brunovistulian terrane acritarchs with Baltica rather than Gondwana, although finally they did not support use of acritarchs to constrain the palaeogeographical position of the Brunovistulian terrane.

Most isotopic ages obtained for plutonic and metamorphic rocks from the Brunovistulian terrane basement are typical of Panafrican (Cadomian) orogenic activity (Van Breemen *et al*. 1982; Finger *et al*. 2000*a*, *b*; Żelaźniewicz *et al*. 2001) although Archaean U–Pb zircon ages were obtained by Bylina *et al*. (2000) from amphibolites in the Upper Silesia Block (Rzeszotary Horst). Cadomian ages are also common for detrital muscovites from the Lower Cambrian clastic sediments of the Brunovistulian terrane (Belka *et al*. 2000). However, Cadomian ages do not permit precise location of the Brunovistulian terrane, as the Cadomian belt was several thousand kilometres long (Nance & Murphy 1994; Torsvik *et al*. 1996). Moreover, the presence of upper Proterozoic basement in the Uralides (Glasmacher *et al*. 1999) indicates that it is not conclusive proof of a Gondwanan origin (Winchester *et al*. 2002). A possible spatial constraint seems to be provided by the Nd model and detrital zircon ages correlated with Grenvillian–Sveconorwegian orogenic activity (Hegner & Kröner 2000; Friedl *et al*. 2001). Ages of about 1.2 Ga have not been noted in the western part of the present northern African margin and Armorican terranes of Western Europe. The Grenvillian–Sveconorwegian orogeny was limited to the Scandinavian margin of Baltica, northern and western margin of South America, and northeastern part of Laurentia (see, e.g. Buchan *et al*. 2000). Therefore some workers locate the Brunovistulian terrane in the South American part of the Cadomian belt (Hegner & Kröner 2000; Friedl *et al*. 2001). It should be emphasized, however, that *c*. 1.15 Ga crust was recently defined also in the Eastern Desert of Egypt (Loizenbauer *et al*. 2001). Moreover, the Grenvillian–Sveconorwegian zircon component was found also in Lower Cambrian sediments of the Małopolska Massif and SW edge of the East European Platform (Belka *et al*. 2002). Thus any palaeogeographical reconstructions based only on isotopic geochronology must be still treated with caution.

Other palaeogeographical models link the Brunovistulian terrane with the Avalonian part of the peri-Gondwanan margin (Moczydłowska 1997; Finger *et al*. 2000*a*) or assume that the Brunovistulian terrane remained near the present SW edge of the East European Craton since the Neoproterozoic (Żelaźniewicz 1998; Żelaźniewicz *et al*. 2001). An Avalonian linkage is, however, in disagreement with palaeomagnetic data. A tropical palaeolatitude for the Brunovistulian terrane in the Early Cambrian shows that it cannot have been a part of Avalonia, which was in higher latitudes on the Gondwana margin at that time.

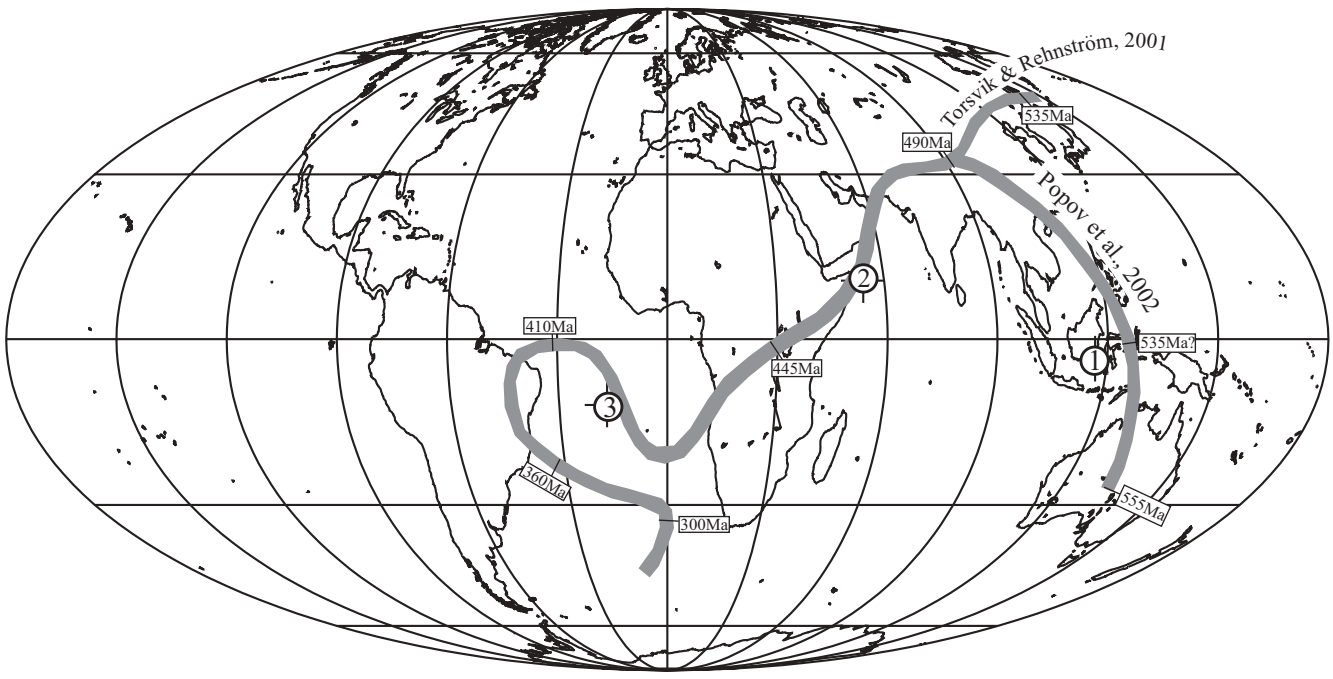


Fig. 7. APWP for Baltica and selected palaeopoles obtained from the Lower Palaeozoic rocks of the Brunovitulian and Małopolska terranes. The Ordovician–Permian segment of the path is according toTorsvik *et al*. (1990, 1996). Two versions of the path are presented for Vendian–Cambrian (see text): 1, palaeopole from this paper; 2, palaeopole from the Middle–Upper Ordovician sediments of the Małopolska terrane (Schätz *et al*. 2002); 3, Late Silurian palaeopole from the Małopolska terrane (Nawrocki 2000).

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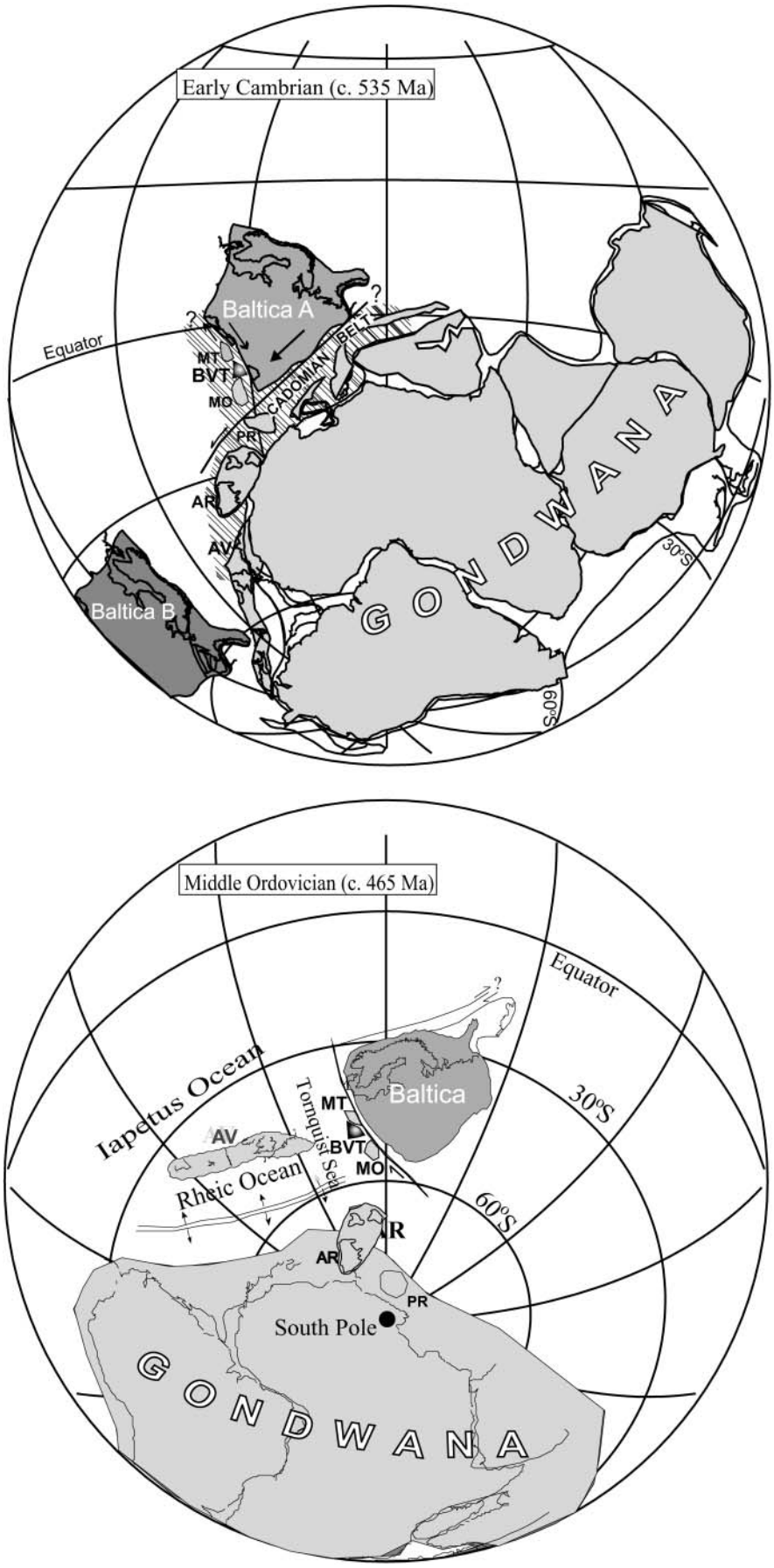


Fig. 8. Simplified palaeogeographical reconstructions showing distribution of selected continents (adopted from Torsvik *et al*. (1996) and Torsvik & Rehnström (2001)) and the position of the Brunovistulian terrane in the Early Cambrian and Mid-Ordovician. Two possible Early Cambrian locations of Baltica are presented (Baltica 'A' is according to the APWP of Popov *et al*. (2002); Baltica 'B' is based on the data of Torsvik & Rehnström (2001)). PR, Perrunica; AV, Avalonia; AR, Armorica; BV, Brunovistulian; MO, Moesia; MT, Małopolska.

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Cambrian latitude estimates for western Avalonia vary between 40 and 50°S (Liss *et al*. 1993). That implies a palaeolatitudinal separation of about 30–40° of the Brunovistulian terrane from the Avalonian margin of Gondwana. The *in situ* development of the Brunovistulian part of Cadomian orogen near the present-day SW edge of the East European Craton could be in agreement with palaeomagnetic data, as then the palaeomagnetic pole from the Brunovistulian terrane is very close to the Popov *et al*. (2002) APWP (Fig. 7, pole 1). However, this solution disagrees with the results of subsidence modelling that point to an extensional regime prevailing along the SW edge of the East European Craton during Vendian–Early Cambrian time (Poprawa *et al*. 1999).

A comparision of our palaeomagnetic data from the Brunovistulian terrane with the Early Cambrian palaeomagnetic data of Torsvik & Rehnström (2001) indicates a palaeolatitudinal separation of about 30° between Baltica and the Brunovistulian terrane (Fig. 8a, Baltica 'B'). In this model the Brunovistulian terrane originally located close to the Arabian part of Gondwana could have been attached to Baltica during the Late Cambrian movement of this continent along the Gondwana margin (Torsvik & Rehnström 2001), followed by migration to its present position on the SW margin when Baltica rotated. This solution, however, does not explain fully the Baltic affinity of some trilobite species from the Brunovistulian terrane.

The Early Cambrian palaeomagnetic record from the Brunovistulian terrane is more consistent with the geological data if the Late Vendian–Cambrian segment of the Baltic APWP is according to Popov *et al*. (2002). The presence of Cadomian basement and the mixed nature of trilobite fauna indicate that in the Early Cambrian the Brunovistulian terrane was located inside the Cadomian belt, between Baltica and Gondwana. Simplified palaeogeographical reconstructions showing possible Early Cambrian and Mid-Ordovician settings of the Brunovistulian terrane are presented in Figure 8. In our model the Brunovistulian terrane occupied a position north of the present-day northern margin of Africa (compare Unrug *et al*. 1999; Leichman & Höck 2001). It was coupled to the present-day southern margin of Baltica during the Cambrian when Baltica moved from equatorial to moderate palaeolatitudes along the Cadomian margin of Gondwana. We cannot exclude, however, that the Brunovistulian terrane could have remained near the present-day southern margin of Baltica since Grenvillian time and was incorporated into the Cadomian orogen later. The Grenvillian zircon component found in the basement of the Brunovistulian terrane (Hegner & Kröner 2000) could support this solution.

The Brunovistulian terrane had to be separated from the peri-Gondwanan region and transported to its present position near the SW edge of the East European Craton before Mid-Ordovician time. It is rather unlikely that the Middle–Upper Ordovician carbonates of the Brunovistulian terrane containing Baltican conodont fauna (Belka *et al*. 2002) were deposited in the polar regions of Gondwana. The tectonic transport could have been related to strike-slip motion along the passive margin of Baltica (Fig. 8; see also Crowley *et al*. 2000). Palaeomagnetic data from the neighbouring Małopolska tectonic block indicate that no large-scale strike-slip activity has occurred along the SW margin of Baltica after Silurian (Nawrocki 2000) or even Mid-Ordovician time (Schätz *et al*. 2002).

Conclusions

Cambrian trilobite fauna from the Brunovistulian terrane are typical for Baltic and Gondwanan zoogeographical provinces and therefore cannot constrain precisely the position of the Brunovistulian terrane during the Early Cambrian. The fauna indicates possible Early Cambrian proximity of the Brunovistulian terrane to both continents. Good quality palaeomagnetic data suggest a nearly equatorial position of the Brunovistulian terrane in the Early Cambrian. Assuming the correctness of the Early Cambrian geographical reconstruction of Gondwana, our results show that at this time the Brunovistulian terrane was distant from the Avalonian margin of Gondwana. It could have been located within the Cadomian belt between Baltica and Gondwana, and presumably occupied a position north of the present-day northern margin of Africa. The Brunovistulian terrane was coupled to the present-day southern margin of Baltica during the Cambrian, when Baltica moved along the Cadomian margin of Gondwana from equatorial to intermediate palaeolatitudes. It cannot be excluded, however, that the Brunovistulian terrane could have remained near the present-day southern margin of Baltica since Grenvillian time and was incorporated into the Cadomian orogen. For both solutions the subsequent dextral tectonic transport of the Brunovistulian terrane along the Baltic margin to its present position is required.

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References

Belka, Z. 1985. Lower Carboniferous conodont biostratigraphy in the northeastern part of the Moravia–Silesia basin. *Acta Geologica Polonica*, **35**, 33–60.

Belka, Z. 1987. The development and decline of a Dinantian carbonate platform: an example from the Moravia–Silesia Basin. *In:* Miller, J., Adams, E.A. & Wright, V.P. (eds) *European Dinantian Environments*. Wiley, Chichester, 177–188.

Belka, Z., Ahrendt, H., Franke, W. & Wemmer, K. 2000. The Baltica–Gondwana suture in central Europe: evidence from K–Ar ages of detrital muscovites and biogeographical data. *In:* Franke, W., Haak, V., Oncken, O. & Tanner, D. (eds) *Orogenic Processes, Quantification and Modelling in the Variscan Belt*. Geological Society, London, Special Publications, **179**, 87–102.

Belka, Z., Valverde-Vaquero, P., Dörr, W., Ahrendt, H., Wemmer, K., Franke, W. & Schäfer, J. 2002. Accretion of first Gondwana-derived terranes at the margin of Baltica. *In:* Winchester, J.A., Pharaoh, T.C. & Verniers, J. (eds) *Palaeozoic Amalgamation of Central Europe*. Geological Society, London, Special Publications, **201**, 19–36.

Bergström, J. 1973. Classification of olenellid trilobites and some Balto-Scandian species. *Norsk Geologisk Tidsskrift*, **53**(3), 283–314.

Biernat, G. & Baliński, A. 1973. Fauna z otworów wiertniczych Sosnowiec IG-1 i Goczałkowice IG-1 (*Stromatoporoidea*, *TaBulata*, *Brachiopoda* i *Trilobita*). *Geological Quarterly*, **17**, 629–630.

Brochwicz-Lewiński, W., Vidal, G., Pożaryski, W., Tomczyk, H. & Zaja.c, R. 1986. Pre-Permian tectonic position of the Upper Silesian Massif (S Poland) in the light of studies on the Cambrian. *Comptes Rendus de l'Académie des Sciences, Série 11*, **303**(16), 1493–1496.

Buchan, K.L., Mertanen, S., Park, R.G., Pesonen, L.J., Elming, S.A., Abrahamsen, N. & Bylund, G. 2000. Comparing the drift of Laurentia and Baltica in the Proterozoic: the importance of key palaeomagnetic poles. *Tectonophysics*, **319**, 167–198.

Buła, Z. 2000. Lower Palaeozoic of Upper Silesia and West Małopolska. *Prace Panstwowego Instytutu Geologicznego*, **171**, 1–89 (in Polish with English summary).

Buła, Z., Jachowicz, M. & Żaba, J. 1997. Principal characteristics of the Upper Silesian Block and Małopolska Block border zone (southern Poland). *Geological Magazine*, **134**(5), 669–677.

Bylina, P., Żelazńiewicz, A. & Dörr, W. 2000. Archean basement in the Upper Silesia Block: U–Pb zircon age from amphibolites of the Rzeszotary Horst. *In: Joint Meeting of EUROPROBE (TESZ) and PACE Projects, Zakopane/Holy Cross Mountains, Poland, Abstracts Volume*. Polish Academy of Science, Polish Geological Institute, Association of Oil & Gas Industrial

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Engineers, Warsaw, 11–12.

Crowley, Q.G., Floyd, P.A., Winchester, J.A., Franke, W. & Holland, G. 2000. Early Palaeozoic rift-related magmatism in Variscan Europe: fragmentation of the Armorican Terrane Assemblage. *Terra Nova*, **12**, 171–180.

Dadlez, R. 1995. Debates about pre-Variscan tectonics of Poland. *Studia Geophysica et Geodaetica*, **39**, 227–234.

Dudek, A. 1980. The crystalline basement block of the outer Carpathians in Moravia: Bruno-Vistulicum. *Rozprawy Československe Akademie Ved*, **90**(8), 81–85.

Finger, F., Hanzl, P., Pin, C., Von Quadt, A. & Steyrer, H.P. 2000a. The Brunovistulian: Avalonian Precambrian sequence at the eastern end of the Central European Variscides? *In:* Franke, W., Haak, V., Oncken, O. & Tanner, D. (eds) *Orogenic Processes: Quantification and Modelling in the Variscan Belt*. Geological Society, London, Special Publications, **179**, 103–112.

Finger, F., Tichomirowa, M., Pin, C. & Hanzl, P. 2000b. Relics of an Early-Panafrican metabasite–metarhyolite formation in the Brno Massif, Moravia, Czech Republic. *International Journal of Earth Sciences*, **89**, 328–335.

Fisher, R.A. 1953. Dispersion on a sphere. *Proceedings of the Royal Society of London, Series A*, **217**, 295–305.

Franke, D. 1995. Caledonian terranes along the southwestern border of the East European Platform—evidence, speculation and open questions. *Studia Geophysica et Geodaetica*, **39**, 241–256.

Friedl, G., Fritz, A., Von Quadt, A., McNaughton, N.J. & Fletcher, I.R. 2001. Results of conventional and SHRIMP U–Pb zircon dating in the south-eastern Bohemian Massif (Austria, Czech Republic): implications for a delineation of different Peri-Gondwana terranes in Variscan Central Europe. *In:* Pharaoh, T.C. & Gee, D. (eds) *ESF EUROPROBE 'Meeting Neoproterozoic–Early Palaeozoic Time-Slice Symposium: Orogeny and Cratonic Response on the Margins of Baltica', Ankara, Abstracts*. Middle East Technical University, Ankara, 16–18.

Geyer, G. 1990. Die marrokanischen Ellipsocephalidae (Trilobita: Redlichiida). *Beringeria*, **3**, 3–363.

Glasmacher, U.A., Giese, U., Stroink, L., Reynolds, P., Alekseyev, A., Puchkov, V. & Bauer, W. 1999. Neoproterozoic terrane at the eastern margin of Baltica—implications for Late Proterozoic palaeogeography and structural evolution of SW Urals, Russia. *Journal of Conference Abstracts*, EUG10. 108.

Gradstein, F.M. & Ogg, J. 1996. A Phanerozoic time scale. *Episodes*, **19**, 1–2.

Hegner, E. & Kröner, A. 2000. Review of Nd isotopic data and xenocrystic and detrital zircon ages from the pre-Variscan basement in the eastern Bohemian Massif: speculations on palinspastic reconstructions. *In:* Franke, W., Haak, V., Oncken, O. & Tanner, D. (eds) *Orogenic Processes, Quantification and Modelling in the Variscan Belt*. Geological Society, London, Special Publications, **179**, 113–130.

Jachowicz, M. & Přichystal, A. 1998. Lower Cambrian sediments in deep boreholes in south Moravia. *Věstnik Českého Geologického Ústavu*, **72**, 329–331.

Jelinek, V. 1977. *The Statistical Theory of Measuring Anisotropy of Magnetic Susceptibility of Rocks and its Application*. Geofyzika, Brno.

Kirschvink, J.L. 1980. The least square line and plane and the analysis of paleomagnetic data. *Geophysical Journal of the Royal Astronomical Society*, **62**, 699–718.

Kirschvink, J.L., Margaritz, M., Ripperdam, R., Zhuravlev, Y. & Rozanov, A. 1991. The Precambrian–Cambrian boundary: magnetostratigraphy and carbon isotopes resolve correlation problems between Siberia, Morocco, and South China. *GSA Today*, **1-4**, 69–91.

Konior, K. & Turnau, E. 1973. Preliminary study of microflora from Lower Devonian deposits in the area of Bielsko-Wadowice. *Rocznik Polskiego Towarzystwa Geologicznego*, **43**, 273–280.

Kotas, A. 1990. Upper Silesian Coal Basin. *In:* Osika, R. (ed.) *Geology of Poland, VI, Mineral Deposits*. Wydawnictwo Geologiczne, Warsaw, 77–92.

Leichman, J. & Höck, V. 2001. The Brunovistulicum: a Gondwana derived terrain accreted to Baltica. *In:* Pharaoh, T.C. & Gee, D. (eds) *ESF EUROPROBE Meeting 'Neoproterozoic–Early Palaeozoic Time-Slice Symposium: Orogeny and Cratonic Response on the Margins of Baltica', Ankara, Abstracts*. Middle East Technical University, Ankara, 37–38.

Liss, M.J., Van der Pluijm, B.A. & Van der Voo, R. 1993. Avalonian proximity of the Ordovician Miramichi Terrane northern New Brunswick, northern Appalachians: paleomagnetic evidence for rifting and back-arc basin formation at the southern margin of Iapetus. *Tectonophysics*, **227**, 17–30.

Loizenbauer, J., Wallbrecher, E., Fritz, H., Neumayr, P., Khudeir, A.A. & Kloetzli, U. 2001. Structural geology, single zircon ages and fluid inclusion studies of the Meatiq metamorphic core complex: implications for Neoproterozoic tectonics in the Eastern Desert of Egypt. *Precambrian Research*, **110**, 357–383.

Lowrie, W. 1990. Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties. *Geophysical Research Letters*, **17**, 159–162.

Moczydłowska, M. 1995. Neoproterozoic and Cambrian successions deposited on the East European Platform and Cadomian basement area in Poland. *Studia Geophysica et Geodaetica*, **39**, 276–285.

Moczydłowska, M. 1997. Proterozoic and Cambrian successions in Upper Silesia: an Avalonian terrane in Southern Poland. *Geological Magazine*, **134**, 679–689.

Nance, R.D. & Murphy, J.B. 1994. Contrasting basement isotopic signatures and the palinspastic restoration of peripheral orogens: example from the Neoproterozoic Avalonian–Cadomian belt. *Geology*, **22**, 617–620.

Narkiewicz, M. 1978. Stratygrafia i rozwój facjalny górnego dewonu mie.dzy Olkuszem a Zawierciem. *Acta Geologica Polonica*, **28**, 415–470.

Nawrocki, J. 2000. Late Silurian paleomagnetic pole from the Holy Cross Mountains: constraints for the post-Caledonian tectonic activity of the Trans-European Suture Zone. *Earth and Planetary Science Letters*, **179**, 325–334.

Orłowski, S. 1975. Lower Cambrian trilobites from Upper Silesia (Goczałkowice borehole). *Acta Geologica Polonica*, **25**, 377–383.

Orłowski, S. 1985. Lower Cambrian and its trilobites in the Holy Cross Mts. *Acta Geologica Polonica*, **35**(3-4), 231–250.

Pharaoh, T.C. 1999. Palaeozoic terranes and their lithospheric boundaries within the Trans-European Suture Zone (TESZ): a review. *Tectonophysics*, **314**, 17–41.

Pisarevsky, S., Gurevich, E. & Khramov, A. 1997. Paleomagnetism of the Lower Cambrian sediments from the Olenek river sections (northern Siberia): paleopoles and the problem of the magnetic polarity in the Early Cambrian. *Geophysical Journal International*, **130**, 746–756.

Popov, V., Iosifidi, A., Khramov, A., Tait, J. & Bachtadse, V. 2002. Paleomagnetism of Upper Vendian sediments from the Winter Coast, White Sea region, Russia: implications for the paleogeography of Baltica during Neoproterozoic times. *Journal of Geophysical Research*, **107**(B11), 1–0 EPM10; 1–8.

Poprawa, P., Šliaupa, S., Stephenson, R.A. & Lazauskiene, J. 1999. Late Vendian–Early Palaeozoic tectonic evolution of the Baltic basin: regional implications from subsidence analysis. *Tectonophysics*, **314**, 219–239.

Pożaryski, W. 1991. The strike-slip terrane model for the North German–Polish Caledonides. *Publications of the Institute of Geophysics, Polish Academy of Sciences*, **A-19**, 3–15.

Schätz, M., Zwing, A., Belka, Z., Soffel, H.C. & Bachtadse, V. 2002. Paleomagnetism of Ordovician carbonate rocks from Małopolska. SE Poland—no evidence of major rotations. EGS General Assembly, Nice. *Geophysical Research Abstracts*, **4**, EGS02-A-03278.

Torsvik, T.H. & Rehnström, E.F. 2001. Cambrian palaeomagnetic data from Baltica: implications for true polar wander and Cambrian palaeogeography. *Journal of the Geological Society, London*, **158**, 321–329.

Torsvik, T.H., Smethurst, M.A., Briden, J.C. & Sturt, B.A. 1990. A review of Paleozoic paleomagnetic data from Europe and their paleogeographic implications. *In:* McKerrow, W.S. & Scotese, C.R. (eds) *Palaeozoic Palaeogeography and Biogeography.* Geological Society, London, Memoirs, **12**, 25–41.

Torsvik, T.H., Smethurst, M.A. & Meert, J.G. *et al.* 1996. Continental break-up and collision in the Neoproterozoic and Paleozoic—a tale of Baltica and Laurentia. *Earth-Science Reviews*, **40**, 229–258.

Unrug, R., Harańczyk, C. & Chocyk-Jamińska, M. 1999. Easternmost Avalonian and Armorican–Cadomian terranes of central Europe and Caledonian–Variscan evolution of the polydeformed Kraków mobile belt: geological constraints. *Tectonophysics*, **302**, 133–157.

Valverde-Vaquero, P., Dörr, W., Belka, Z., Franke, W., Wiszniewska, J. & Schastok, J. 2000. U–Pb single-grain dating of detrital zircon in the Cambrian of central Poland: implications for Gondwana versus Baltica provenance studies. *Earth and Planetary Science Letters*, **184**, 225–240.

Van Breemen, O., Aftalion, M., Bowes, D.R., Dudek, A., Misar, Z., Povondra, P. & Vrana, S. 1982. Geochronological studies of the Bohemian Massif, Czechoslovakia, and their significance in the evolution of Central Europe. *Transactions of the Royal Society of Edinburgh: Earth Sciences*, **73**, 89–108.

Winchester, J.A. & the PACE TMR NetworkTeam 2002. Paleozoic amalgamation of Central Europe: new results from recent geological and geophysical investigations. *Tectonophysics*, **360**, 5–21.

Żaba, J. 1999. The structural evolution of Lower Palaeozoic succession in the Upper Silesia Block and Małopolska Block border zone (Southern Poland). *Prace Panstwowego Instytutu Geologicznego*, **166**, 1–162 (in Polish with English summary).

Żelaźniewicz, A. 1998. Rodinian–Baltican link of the Neoproterozoic orogen in southern Poland. *Acta Universitatis Carolinae, Geologica*, **42**, 509–511.

Żelaźniewicz, A., Seghedi, A., Jachowicz, M., Bobiński, W., Buła, Z. & Cwojdziński, S. 2001. U–Pb SHRIMP data confirm the presence of a Vendian foreland flysch basin next to the East European Craton. *In:* Pharaoh, T.C. & Gee, D. (eds) *ESF EUROPROBE Meeting 'Neoproterozoic–Early Palaeozoic Time-Slice Symposium: Orogeny and Cratonic*

Downloaded from http://jgs.lyellcollection.org/ at University of California-San Diego on March 4, 2017

Response on the Margins of Baltica', Ankara, Abstracts. Middle East Technical University, Ankara, 98–100.

ZIJDERVELD, J.D.A. 1967. AC demagnetization of rocks: analysis of results. *In:* COLLINSON, D.W., CREER, K.M. & RUNCORN, S.K. (eds) *Methods in Paleomagnetism*. Elsevier, New York, 254–287.

ŻYLIŃSKA, A. 2002. Stratigraphic and biogeographic significance of Late Cambrian trilobites from Lysogóry (Holy Cross Mountains, central Poland). *Acta Geologica Polonica*, **52**, 217–238.

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